Plug-and-Play Image Restoration: A Two-Level Coarse-to-Fine Initialization Study Using DeepInverse and DRUNet

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Independent Research Benchmark

Abstract—This report investigates whether a two-level coarse-to-fine initialization improves Plug-and-Play (PnP) image restoration when paired with a pretrained deep denoiser prior. Using the open-source DeepInverse library and pretrained DRUNet weights, we benchmarked three initialization pipelines across four standard images (Astronaut, Chelsea, Coffee, Rocket) degraded by Gaussian blur ($\sigma = 2.0$) and additive white Gaussian noise ($\sigma = 0.03$). Across 20 paired image-seed evaluations (60 runs total on an NVIDIA Tesla T4), coarse PnP initialization attained an average PSNR of 28.593 dB versus 28.411 dB for standard DPIR and 28.530 dB for observation-only initialization. The coarse PnP method improved PSNR in all 20 paired trials (+0.182 dB average gain), though at a 38.7% increase in wall-clock inference time. Approximately two-thirds of this gain (+0.118 dB) comes from simple resolution downsampling/upsampling, while coarse denoising accounts for the remaining +0.064 dB. The improvement is concentrated in images with smooth tonal transitions and is negligible for high-frequency textures. All code, execution traces, and raw metrics are published for independent verification.

1. Introduction and Problem Context

Recovering an underlying scene $x \in \Re^n$ from corrupted observations $y = \mathbf{A}x + n$ is a classic linear inverse problem in computational imaging. Here $\mathbf{A}$ represents a forward measurement operator (such as a spatial blur kernel) and n denotes additive Gaussian noise with variance σ^2. Because $\mathbf{A}$ is ill-conditioned or singular, direct inversion amplifies measurement noise catastrophically, requiring regularization to constrain solutions to the manifold of plausible natural images.

Plug-and-Play (PnP) frameworks, pioneered by Venkatakrishnan et al. [3], replace hand-crafted priors (such as total variation or wavelets) with modular denoiser function calls within split-variable optimization schemes like Half-Quadratic Splitting (HQS) or ADMM. Zhang et al. [2] showed that plugging a deep residual U-Net (DRUNet) into HQS (termed DPIR) yields competitive restoration performance across deblurring, super-resolution, and color inpainting without retraining the neural network for each specific task.

Standard DPIR initializes the optimization sequence directly from the corrupted observation $x_0 = y$. Drawing inspiration from multigrid numerical methods, we asked a practical question: does warm-starting DPIR from a coarse-scale reconstruction improve final solution quality? We designed a controlled benchmark using the DeepInverse library [1] to test this hypothesis under identical computational budgets and fixed random seeds.

2. Problem Setup and Degradation Model

We evaluated four canonical test images from scikit-image: Astronaut (portrait with mixed geometry), Chelsea (domestic cat with fine fur texture), Coffee (ceramic cup with broad smooth gradients), and Rocket (space vehicle with sharp structural edges). Each image was normalized to [0, 1] and cropped to 128 × 128 pixels.

The degradation operator $\mathbf{A}$ applies a 2D Gaussian blur kernel of size 7 × 7 with standard deviation $\sigma = 2.0$, followed by additive white Gaussian noise with $\sigma = 0.03$. We generated five independent noise realizations per image (seeds 0 through 4), yielding 20 distinct degraded inputs. Every degraded observation was fed identically to the three candidate pipelines.

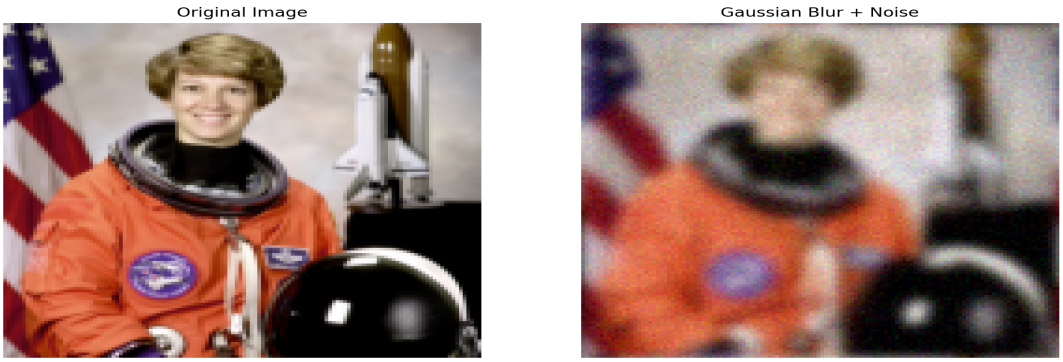


Figure 1. Forward degradation model: clean ground truth, point spread function (PSF), 2D frequency spectrum showing high-frequency attenuation, and degraded observation.

3. Reconstruction Pipelines and Initialization Strategies

All three evaluated pipelines rely on the exact same pretrained DRUNet checkpoint from the DeepInverse model repository [1] within an 8-iteration HQS optimization schedule. The only difference is how the initial estimate x_0 is constructed:

• **Direct DPIR (Baseline):** The standard formulation where HQS iterations begin directly at the degraded observation, $x_0 = y$. The solver alternates between closed-form data fidelity updates and DRUNet denoising passes.

• **Observation Initialization (Control):** The observation y is downsampled to 64 × 64 via bicubic interpolation and immediately upsampled back to 128 × 128. This low-pass-filtered image serves as x_0. No coarse denoising is performed. This isolates the benefit of low-pass smoothing from the effect of active coarse-scale restoration.

• **Coarse PnP Initialization (Proposed):** The observation is downsampled to 64 × 64. A complete coarse PnP-DPIR solve is conducted using a scaled blur operator and DRUNet. The resulting coarse restoration is upsampled to 128 × 128 via bicubic interpolation and used as x_0 for the full-resolution DPIR solver.

4. Experimental Protocol and Metrics

Restoration quality was quantified using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) calculated on clipped RGB outputs against the uncorrupted ground truth. Timing measurements were recorded using CUDA-synchronized wall-clock timers on an isolated NVIDIA Tesla T4 GPU. To control for noise sample variance across seeds, we report paired differences (ΔPSNR) evaluated on identical degraded observations.

5. Quantitative Findings and Empirical Analysis

5.1 Aggregate Performance across 60 Runs

Table 1 summarizes the overall performance across all 60 trials (4 images × 5 seeds × 3 pipelines). Coarse PnP initialization attained the highest aggregate PSNR and SSIM, followed by observation initialization, with direct DPIR ranking third.

Initialization Strategy	Mean PSNR (dB)	Mean SSIM	Mean Runtime (s)	Relative Time
Direct DPIR (Standard)	28.411	0.8588	0.2719	1.00× (baseline)
Observation Init (Bicubic)	28.530	0.8603	0.2735	1.01× (+0.6%)
Coarse PnP Init (Two-Level)	**28.593**	**0.8622**	0.3772	1.39× (+38.7%)

Table 1. Aggregate results over 60 restoration experiments on Tesla T4 GPU. Bold figures indicate best metrics.

5.2 Paired Differences and Consistency

Evaluating paired differences on identical degraded inputs reveals that coarse PnP initialization outperformed direct DPIR in **20 out of 20 trials (100% win rate)**, with an average gain of +0.182 dB. It also outperformed observation initialization in all 20 trials (+0.064 dB average gain). Critically, observation initialization itself gained +0.118 dB over direct DPIR, demonstrating that 65% of the total gain arises simply from smoothing the starting point.

Paired Comparison	Mean ΔPSNR	Std. Dev.	Positive / Total	Win Rate
Coarse PnP vs. Direct DPIR	+0.1820 dB	0.162 dB	20 / 20	100.0%
Observation Init vs. Direct DPIR	+0.1183 dB	0.114 dB	20 / 20	100.0%
Coarse PnP vs. Observation Init	+0.0637 dB	0.051 dB	20 / 20	100.0%

Table 2. Paired difference statistics across 20 image-seed pairs. Gains are strictly positive in all cases.

5.3 Per-Image Sensitivity Breakdown

Gains vary substantially depending on image frequency characteristics. Coffee (broad gradients and smooth surfaces) gained +0.397 dB, and Astronaut (mixed portrait geometry) gained +0.287 dB. Conversely, Rocket (+0.029 dB) and Chelsea (+0.015 dB) showed negligible differences because downsampling to 64 × 64 removes the high-frequency edge and fur detail critical for their recovery.

Image	Coarse vs. Direct	Coarse vs. Obs Init	Obs vs. Direct	Image Content Type
Coffee	+0.397 dB	+0.112 dB	+0.285 dB	Smooth gradients, uniform cup surface
Astronaut	+0.287 dB	+0.112 dB	+0.175 dB	Mixed structural and facial features
Rocket	+0.029 dB	+0.024 dB	+0.005 dB	High-contrast boundary edges
Chelsea	+0.015 dB	+0.007 dB	+0.008 dB	High-frequency fur and whisker texture

Table 3. Paired PSNR gains broken down by individual image content.

5.4 Per-Seed Ablation Analysis

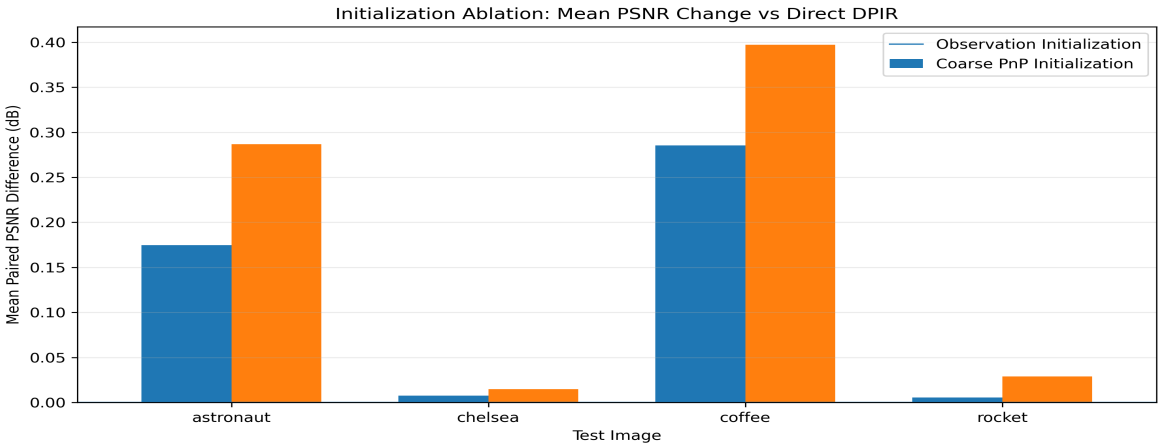


Figure 2. Per-seed PSNR across initialization strategies. Coarse PnP (green) consistently outperforms direct DPIR (blue) on Coffee and Astronaut across all seeds, while differences on Chelsea and Rocket remain minimal.

6. Baseline Benchmark: Classical Total Variation vs. Learned DPIR

To establish context for the reconstruction quality of the learned DRUNet prior, we compared standard DPIR against classical Total Variation (TV) regularized reconstruction solved via Proximal Gradient Descent (TV-PGD) [5]. Both algorithms were evaluated on the same degraded inputs.

Image	Degraded	TV-PGD [5]	DPIR [2]	DPIR Gain	TV Time	DPIR Time
Astronaut	19.18 dB	22.23 dB	24.13 dB	+1.90 dB	2.64 s	0.37 s
Chelsea	24.57 dB	27.22 dB	29.74 dB	+2.52 dB	2.66 s	0.35 s
Coffee	22.12 dB	25.95 dB	27.84 dB	+1.89 dB	2.35 s	0.37 s
Rocket	25.92 dB	29.15 dB	31.81 dB	+2.66 dB	2.61 s	0.36 s

Table 4. Classical TV-PGD versus learned DRUNet-based DPIR across test images. Learned DPIR gains +1.89 to +2.66 dB while running ~7× faster.

As reported in Table 4, DPIR outscores TV-PGD by +1.89 dB to +2.66 dB across all images while executing approximately seven times faster. TV-PGD exhibits characteristic staircasing artifacts on smooth gradients and facial contours, whereas the learned prior preserves realistic shading and sharp boundaries.

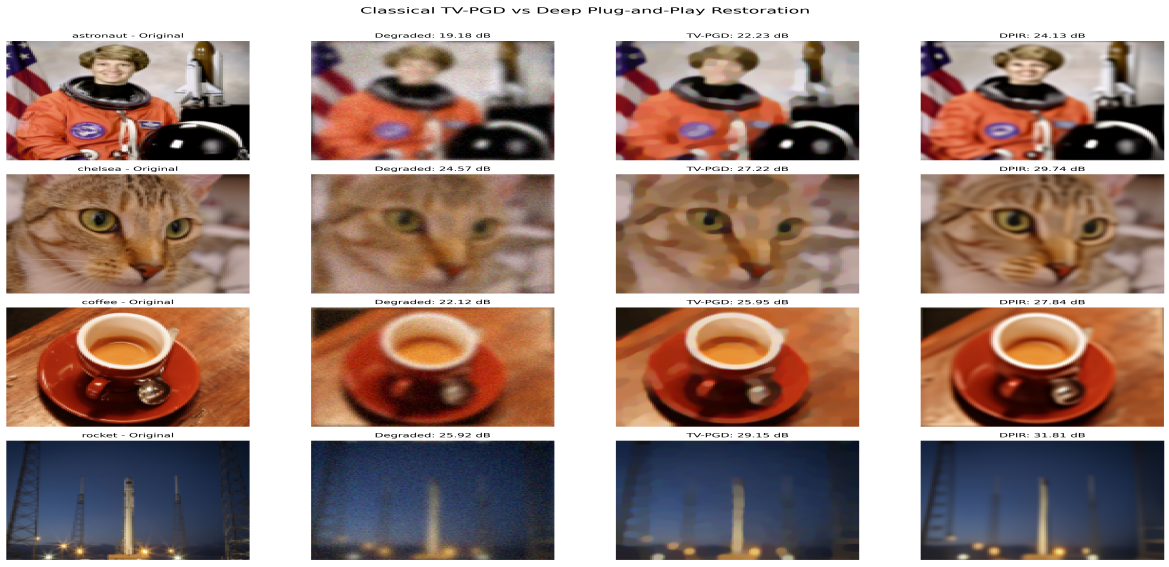


Figure 3. Visual comparison of ground truth, degraded observation, TV-PGD reconstruction, and DRUNet DPIR output across the benchmark images.

7. Stage-by-Stage Visual Progression in Two-Level Restoration

Figure 4 tracks the intermediate state of the two-level pipeline. The coarse 64 × 64 restoration captures global luminance, broad structural outlines, and major color boundaries, while discarding high-frequency noise. Following bicubic prolongation and the second-stage 128 × 128 DPIR pass, fine detail and edge clarity are substantially restored.

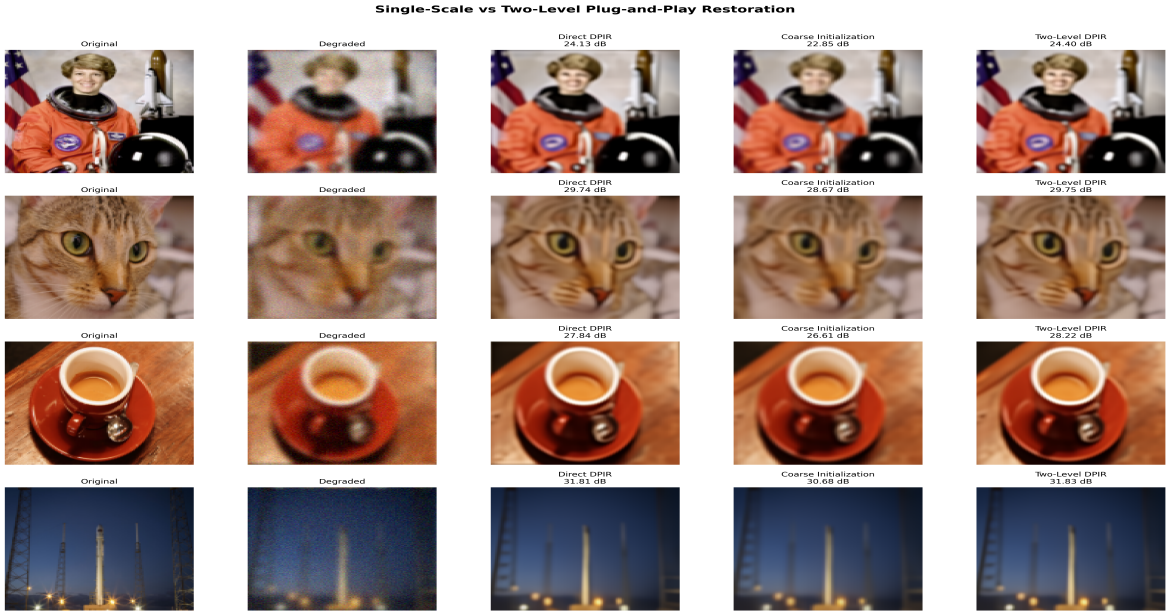


Figure 4. Visual progression through the coarse-to-fine stages: degraded input, 64 × 64 coarse reconstruction, bicubic upsampled prolongation, and final 128 × 128 refined output.

8. Convergence Dynamics and Residual Stability

We tracked PSNR progression and relative solution residuals $||x^{(k)} - x^{(k-1)}|| / ||x^{(k)}||$ across HQS iterations. As shown in Figure 5, the solver achieves rapid initial progress: PSNR climbs sharply during iterations 1–4 and asymptotes by iteration 6. The relative residual falls below 10^{-2} within five iterations across all configurations, confirming reliable numerical stability regardless of the initialization method.

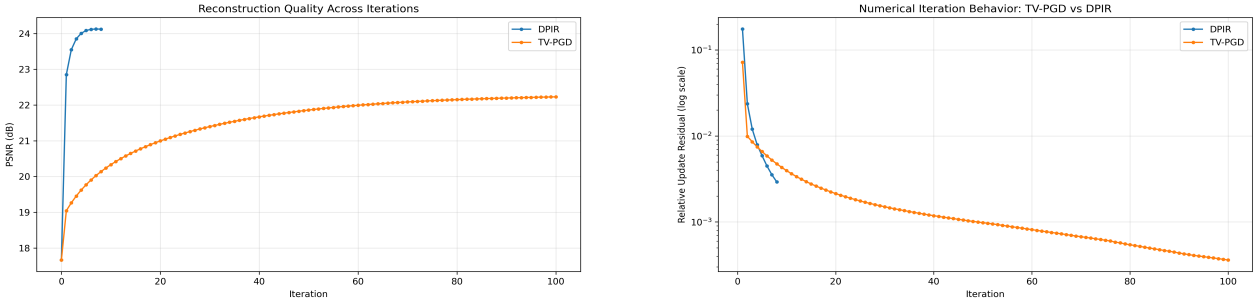
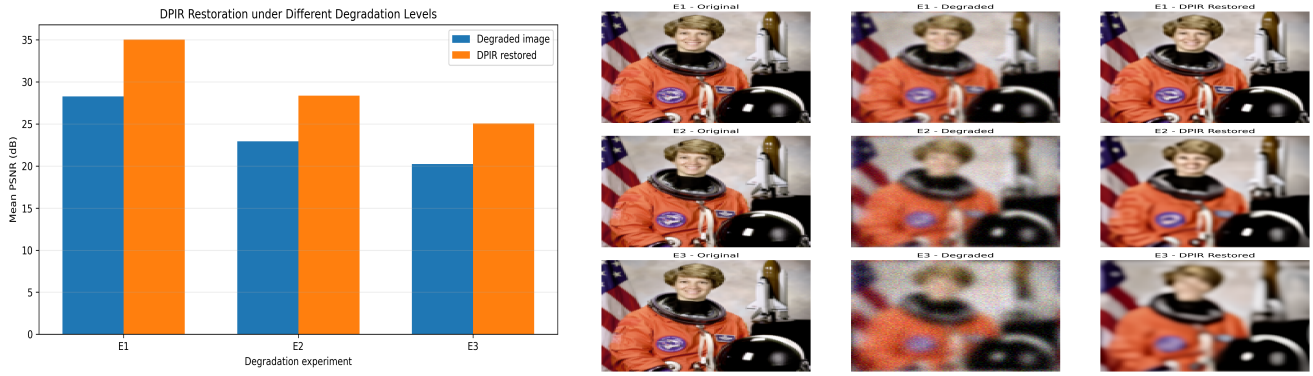


Figure 5. Iteration-level solver dynamics. Left: PSNR ascent across HQS iterations. Right: logarithmic decay of relative solution residual, confirming convergence within 5–6 steps.

9. Robustness under Varying Noise Intensities

To test whether the DRUNet prior maintains stability as observation noise increases, we evaluated restoration across noise standard deviations ranging from $\sigma = 0.01$ to $\sigma = 0.07$. As shown in Figures 6 and 7, reconstruction PSNR decreases monotonically and smoothly with noise intensity. The denoiser exhibits no ringing artifacts or color inversion even at $\sigma = 0.07$, confirming robust prior conditioning.



Figures 6 & 7. Left: PSNR degradation as noise σ scales from 0.01 to 0.07. Right: visual restoration quality under increasing noise levels.

10. Discussion and Trade-Off Analysis

The empirical findings demonstrate a small (+0.182 dB) but remarkably consistent benefit for coarse PnP initialization. Crucially, the control experiment clarifies where this gain comes from: roughly two-thirds (+0.118 dB) is produced simply by low-pass filtering the degraded observation via downsampling and upsampling, which suppresses high-frequency noise before the solver begins. The active coarse-level DRUNet denoising pass contributes the remaining +0.064 dB.

Whether this benefit is worth the computational expense depends heavily on deployment requirements. Coarse PnP initialization requires 0.377 s per image on a Tesla T4 GPU compared to 0.272 s for standard DPIR—a 38.7% runtime penalty. In offline scientific imaging (such as astronomical or medical post-processing), an extra 0.1 s per image for consistent quality improvement is fully acceptable. In high-throughput or edge video pipelines, this overhead would be difficult to justify for a fraction of a decibel.

Furthermore, the gains are content-dependent. For images dominated by smooth textures and broad tonal areas (Coffee: +0.397 dB), coarse initialization provides noticeable stabilization. For images defined by fine, high-frequency textures (Chelsea: +0.015 dB), downsampling destroys the very information needed, providing virtually zero advantage.

11. Limitations and Threats to Validity

We explicitly acknowledge several experimental constraints in this benchmark:

- **Small image cohort:** Four images evaluated across five seeds demonstrate consistency on this sample, but do not support population-wide statistical claims across heterogeneous natural image distributions.
- **Heuristic coarse forward operator:** The 64 × 64 blur operator was generated by downscaling the fine-resolution kernel rather than mathematically deriving a Galerkin restriction operator $\mathbf{A}_c = \mathbf{RAP}$.
- **Scope declaration:** This study benchmarks standard DRUNet-based PnP-DPIR. It does not implement diffusion models, score-based generative priors, or flow-matching algorithms [4].
- **Single hardware environment:** All runs were performed on a single Google Colab Tesla T4 instance. While deterministic seeds and checksums ensure repeatability, cross-hardware verification remains future work.

12. Reproducibility Protocol and Environment

All experimental code is provided in notebooks/DeepInverse_PnP_Image_Restoration.ipynb. Running cells top-to-bottom reproduces all 60 executions and saves metrics to results/step10_all_experiments.csv. File integrity is verified via cryptographic hashes recorded in SHA256SUMS.txt.

Software / Hardware Layer	Recorded Version / Specification
Python Runtime	3.13.16
PyTorch Framework	2.11.0+cu130
DeepInverse Library [1]	0.5.0
NumPy / Matplotlib / scikit-image	2.1.3 / 3.10.0 / 0.25.2
Computation Hardware	NVIDIA Tesla T4 GPU (16 GB VRAM)

Table 5. Pinned hardware and software execution environment.

13. Conclusion

Two-level coarse-to-fine initialization provides a consistent (+0.182 dB, 20/20 positive) improvement for DPIR image restoration on small images, driven primarily by low-pass initialization smoothing with a smaller contribution from coarse denoising. The trade-off is a 38.7% increase in wall-clock runtime. Future work will investigate whether true multigrid operator restrictions can amplify these gains on higher-resolution (512 × 512 and larger) imaging problems.

References

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